CS177 Bioinformatics: Software Development and Applications
SIST, ShanghaiTech University

Spring, 2025

Homework 1

Due time: 11:59 pm, March 29, 2025 (Beijing Time).
Submission: In Blackboard, upload your solution in a zip file, including a PDF file with the written solution and Jupyter notebooks, etc.
Note: Please name your zip file as "CS177-HW1-your name-your student ID", for example, "CS177-HW1-JieZheng-2024233187.zip".

These questions below are assigned 50 points in total, which will constitute 10% of the final grade.

Q1. Probability (15 points)

1) Assume a randomly generated DNA sequence of length n, where each position's base (i.e. A, T, G, or C) is independently and uniformly chosen. If a random DNA sequence of length 100,000 is generated, what is the expected number of occurrences of the restriction enzyme EcoRI's recognition site "GAATTC"? (5 points)

2) You are given an urn containing 10 balls, each of which is either white or black. You are told that the number of white balls in the urn is equally likely to be any number between 0 and 10 inclusive. Thus in particular the probability that all the 10 balls are white is $\frac{1}{11}$.

   (a) You pick a ball uniformly at random and it is white. Given this event, what is now the probability that all the 10 balls are white? (5 points)
   Hint: Use Bayes' Rule:
   $$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

   (b) You pick k balls at random **with replacement** and all are white. What is now the probability that all the 10 balls are white, as a function of k? (5 points)

Q2. Hidden Markov Model (20 points)

Consider a Hidden Markov Model (HMM) $\lambda = (Q, V, p, A, E)$ with the following parameters:

- **State set:** $Q = \{q_1, q_2, q_3\}$ (three hidden states).
- **Observation symbol set:** $V = \{v_1, v_2, v_3\}$.
- **Transition probability matrix** A**:**

$$A = \begin{bmatrix} \frac{2}{5} & \frac{3}{10} & \frac{3}{10} \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

- **Initial state probabilities:** $\mathbf{p} = [1, 0, 0]$, meaning $p_1 = 1$, $p_2 = 0$, $p_3 = 0$.

- **Emission probability matrix E:**

$$E = \begin{bmatrix} \frac{3}{10} & \frac{3}{10} & \frac{2}{5} \\ \frac{1}{5} & 0 & \frac{4}{5} \\ 0 & \frac{3}{5} & \frac{2}{5} \end{bmatrix}$$

The element in the i-th row and j-th column of matrix E represents the probability of emitting observation symbol o_j from state s_i, denoted as $P(o_j|s_i)$.

(a) What are all the possible state paths for the observed sequence $V = (v_1, v_2, v_3)$? (5 points)

(b) Manually execute the forward algorithm to calculate the likelihood $P(V|\lambda)$. (5 points)

Note: You should write each step of derivation in the calculation process. No score will be given if you write only the final result.

(c) Complete the Python code in the enclosed file `hmm.ipynb` for calculating the likelihood $P(V|\lambda)$. (10 points)

Q3. HPC Test (5 points)

For this assignment, you need to set up and use the ShanghaiTech HPC Cluster with GPU support. If you attended the TA class tutorial, you may have already completed some of these steps. If not, you can follow the tutorial available at this link.

Important: This assignment must be completed on the ShanghaiTech HPC Cluster (using Slurm or PBS), not on your personal computer.

Tasks:

1. Log into the HPC Cluster.
2. Use the `module` command to load the Anaconda environment.
3. Create a new environment and install PyTorch correctly.
4. Request GPU resources using a command similar to:

```
salloc --partition=CS177 --nodes=1 --ntasks=1 --cpus-per-task=32
--gres=gpu:NVIDIAGeForceGTX1080:1 --time=24:00:00 --mem=64G
```

   Use srun to access the allocated node.

5. Use the command below to test your environment. You should **include** the screenshot of the outcome of running the following command:

```
nvidia-smi
```

   *(Tip: you should activate your **conda environment** before you run the following commands.)*

```
python
```

   (In Python, run:)

```
import torch
torch.cuda.is_available()
```

   Grading Criteria (5 points total):

   - Successfully logging into the HPC cluster.

- Correctly loading the Anaconda environment and installing PyTorch.
- Successfully requesting GPU resources using Slurm.
- Running `nvidia-smi` and providing a correct screenshot of the GPU status.
- Running the Python script to check CUDA availability and providing a correct screenshot.

Example of expected screenshots:

```
(cs177) [hongmt2022-cs177@sist_gpu53 ~]$ nvidia-smi
Wed Mar 12 13:40:22 2025
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 550.54.15              Driver Version: 550.54.15      CUDA Version: 12.4     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA GeForce GTX 1080        On  |   00000000:02:00.0 Off |                  N/A |
| 38%   54C    P0             55W /  180W |      65MiB /   8192MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
```

```
(cs177) [hongmt2022-cs177@sist_gpu53 ~]$ python
Python 3.10.16 (main, Dec 11 2024, 16:24:50) [GCC 11.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import torch
>>> torch.cuda.is_available()
True
```

Q4. Deep Learning Model Building (10 points)

Objective: Implement a CNN model for breast cancer image classification.

Problem Description: A dataset containing breast cancer images is provided. You need to complete the Python code in the attached Jupyter notebook (`BreastCancerClassification.ipynb`) to implement a CNN model and evaluate its performance on the test set. Follow the instructions given in the Jupyter notebook.

The score is based 50% on code correctness, ensuring the implementation runs without errors and follows the given instructions, and 50% on model performance, where achieving at least 80% accuracy earns full points.

END OF CS177 HOMEWORK 1